

Software Engineering Project

Jan 2026 Term - Team 051

Milestone - 2



Submitted by:

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Design of Components

1. Customer View (Ordering Interface / AI Interaction)

- Scan QR code to access the restaurant ordering interface.
- Browse the digital menu categorized into appetizers, main course, desserts, and beverages.
- Search dishes using the search bar.
- Add menu items to the cart and adjust quantities before ordering.
- Use the AI Smart Assistant to create a custom dish based on dietary preferences such as high-protein, low-calorie, vegetarian, or other custom requirements.
- Send the order to the system, which is forwarded to the manager for approval before reaching the kitchen.
- View the order status while waiting for manager approval.
- View order history and reorder previous meals if needed.

2. Manager View (Restaurant Operations & Approval System)

- View and manage incoming customer orders before they are sent to the kitchen.
- Approve or reject customer orders to prevent fake or unnecessary orders.
- Manage the restaurant menu, including adding new dishes and updating prices.
- Set fixed pricing for standard menu items and define profit margins for dynamically priced AI-generated custom dishes based on ingredient cost.
- Manage inventory items and track ingredient availability.
- The manager dashboard contains three QR generation buttons for:
 - Chef
 - Waiter
 - Customer tables
- When the manager clicks on QR generation, the system asks for the number of tables for which QR codes need to be generated.
- The system then generates QR codes for each table which customers can scan to place orders.
- The manager can also access and monitor the Chef Dashboard and Waiter Dashboard from the manager panel.
- The inventory system highlights low stock items using a different background color alert to notify the manager to restock ingredients.
- View restaurant performance metrics, including total orders and revenue.

- Access reports and analytics to understand business performance.

3. Chef / Kitchen View (Kitchen Dashboard)

- View incoming approved orders in a structured table format.
- Orders appear in a priority queue depending on category or urgency.
- Different background colors indicate the dish category.
- View order details including table number, dish name, quantity, and category.
- For AI-generated custom dishes, the chef can use the View button to see the recipe and ingredients.
- Mark dishes as prepared once cooking is complete.

4. Waiter View (Order Delivery & Payment Interface)

- The waiter dashboard displays all restaurant tables in a list format such as Table 1, Table 2, Table 3, etc.
- When the waiter selects a specific table, the system shows all orders associated with that table.
- The waiter can serve the prepared dishes to the customer.
- After serving, the waiter can proceed to the payment section.
- Customers can complete the payment using:
 - QR-based online payment
 - Cash payment
- Once the payment is completed, the waiter can close the order and mark that the table order has been finished.

5. Inventory Management Component

- Track the availability of ingredients used in dishes.
- Automatically update stock levels after an order is approved.
- Prevent orders for dishes whose ingredients are unavailable.
- Help the manager monitor ingredient usage and inventory levels.

6. AI Recommendation Component

- Accept user input describing a custom dish.
- Analyze dietary preferences and available ingredients.
- Generate a suitable dish suggestion with preparation details.
- Send the AI-generated dish to the system as a custom order.

7. Notification Component

- Notify the manager when a new order is submitted.
- Notify the chef when an order is approved.
- Notify the waiter when a dish is ready for delivery.
- Provide real-time updates for order status.

Class Diagram

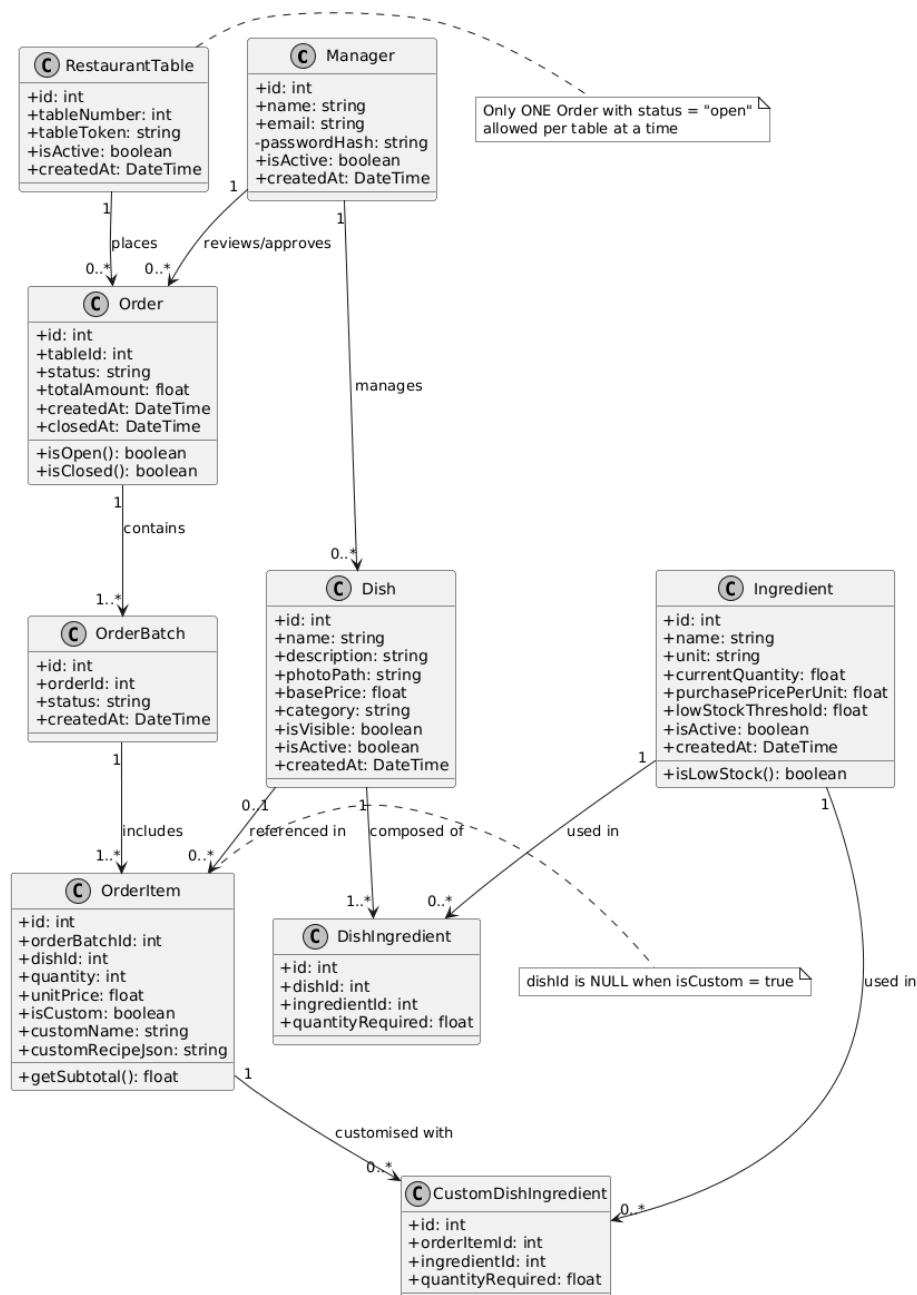


Image 1. Class Diagram

Databased Design diagram

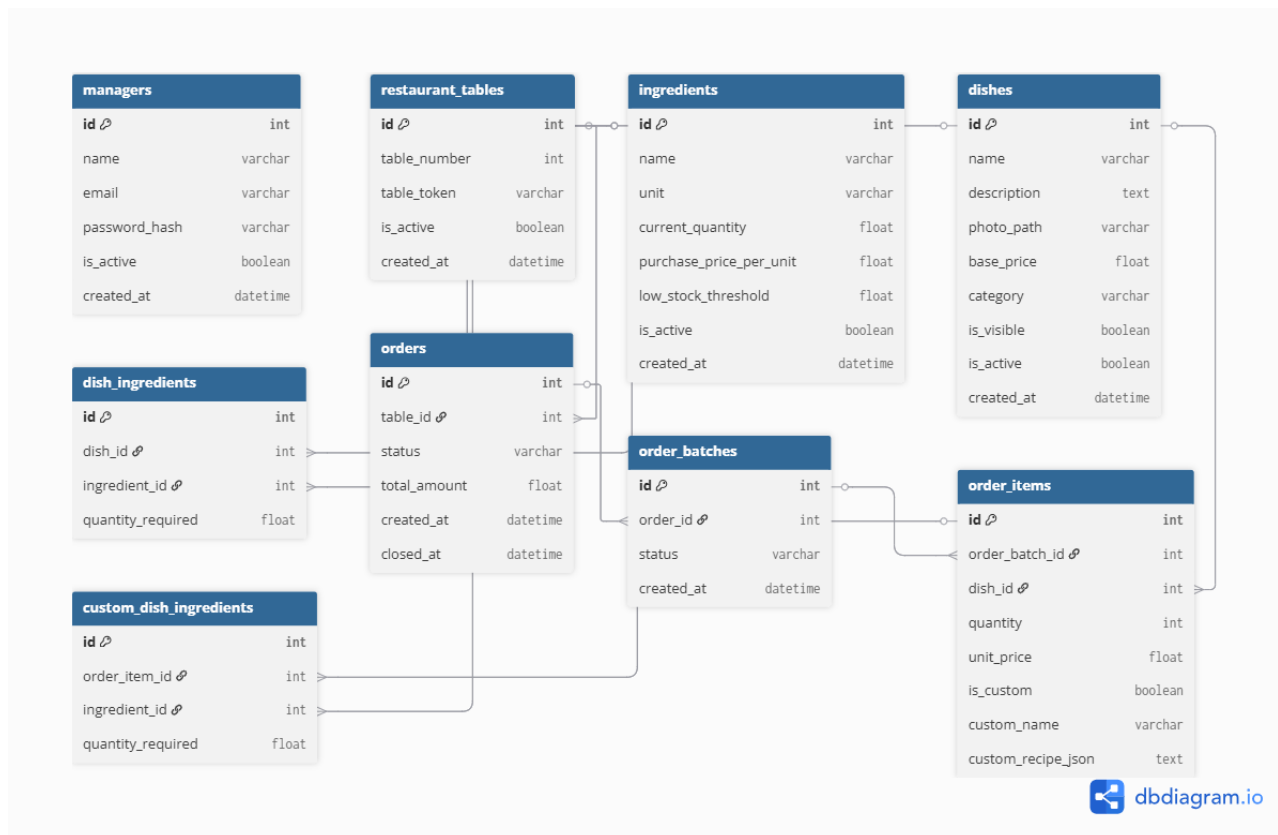


Image 2. ER Diagram

Sprint Schedule

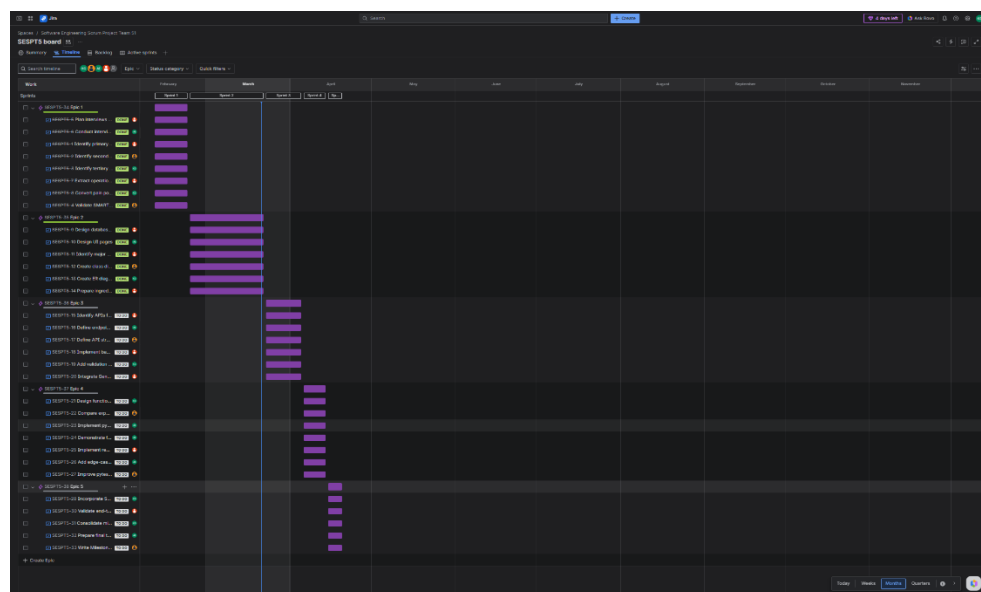


Image 3. Sprint Schedule TimeLine

More Sprint Schedule TimeLine Screen Shots can be found here – [Link](#)

SCRUM Meetings Schedule and Minutes

(SCRUM Meetings: Conducted regularly by the team during project development.)

Minutes from Scrum Meeting 1:

In this meeting, we identified the different types of users involved in the system, namely primary users, secondary users, and tertiary users. We also discussed and finalized the type of business we will be working on. Additionally, we planned the interviews by deciding which team member will take interviews and for which businesses.

Minutes from Scrum Meeting 2:

After conducting the interviews, we discussed the findings and distributed the work among team members. One member was assigned to focus on primary users and write their user stories, another member handled secondary users, and the third member worked on tertiary user stories. We also discussed the pain points identified during the interviews.

Minutes from Scrum Meeting 3:

In this meeting, each team member presented their user stories along with the identified pain points. We discussed how to convert these into SMART user stories as per the guidelines. We also explored possible solutions to address the pain points of all three types of users. Additionally, we discussed the type of system to be developed (web application or mobile application) and the basic structure of the system. We also discussed how to prepare and submit the Milestone 1 report.

Minutes from Scrum Meeting 4:

We created the initial database for the system, which was prepared by one team member and reviewed by the team during the meeting. After that, we distributed the work for UI design, where each team member was assigned to design the UI for primary, secondary, and tertiary users respectively.

Minutes from Scrum Meeting 5:

In this meeting, we presented our work to the business owner and collected feedback regarding the design and functionality. Based on the feedback, we planned the next steps. We started working on the frontend of the web application. We also ensured that all tasks and progress were regularly updated in the Jira application for proper tracking and management.

Minutes from Scrum Meeting 6:

We reviewed the frontend implementation and discussed possible improvements. We also took feedback again from the business owner, who confirmed that the work was aligned with expectations. In this meeting, we planned how to proceed with backend development and discussed the ingredients database prepared by one team member. We also discussed how to prepare the Milestone 2 report, including design of components, ER diagram, and class diagram, and assigned responsibilities for these tasks.

Screenshot of Gantt Chart

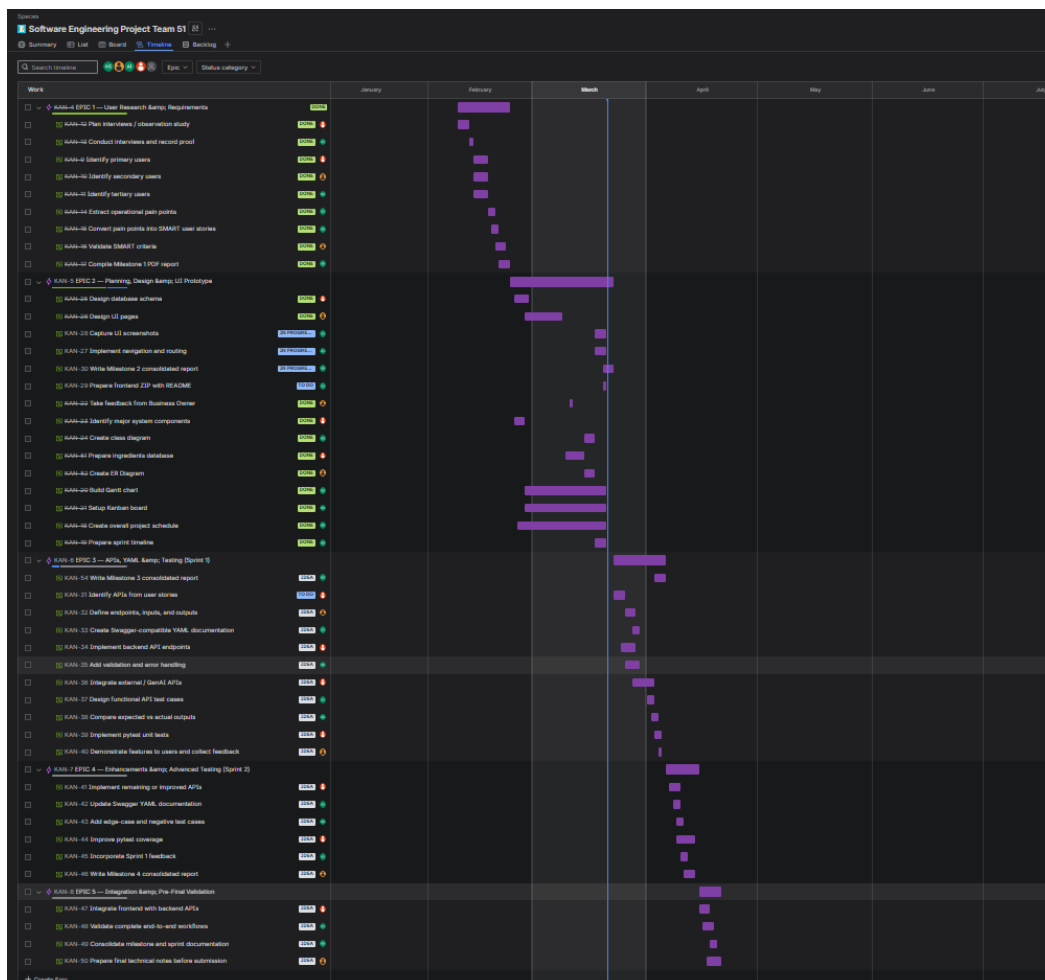


Image 4. Gantt Chart

More Clear Screen Shots of Gantt Chart can be found here – [Link](#)

Screenshot of Kanban Board

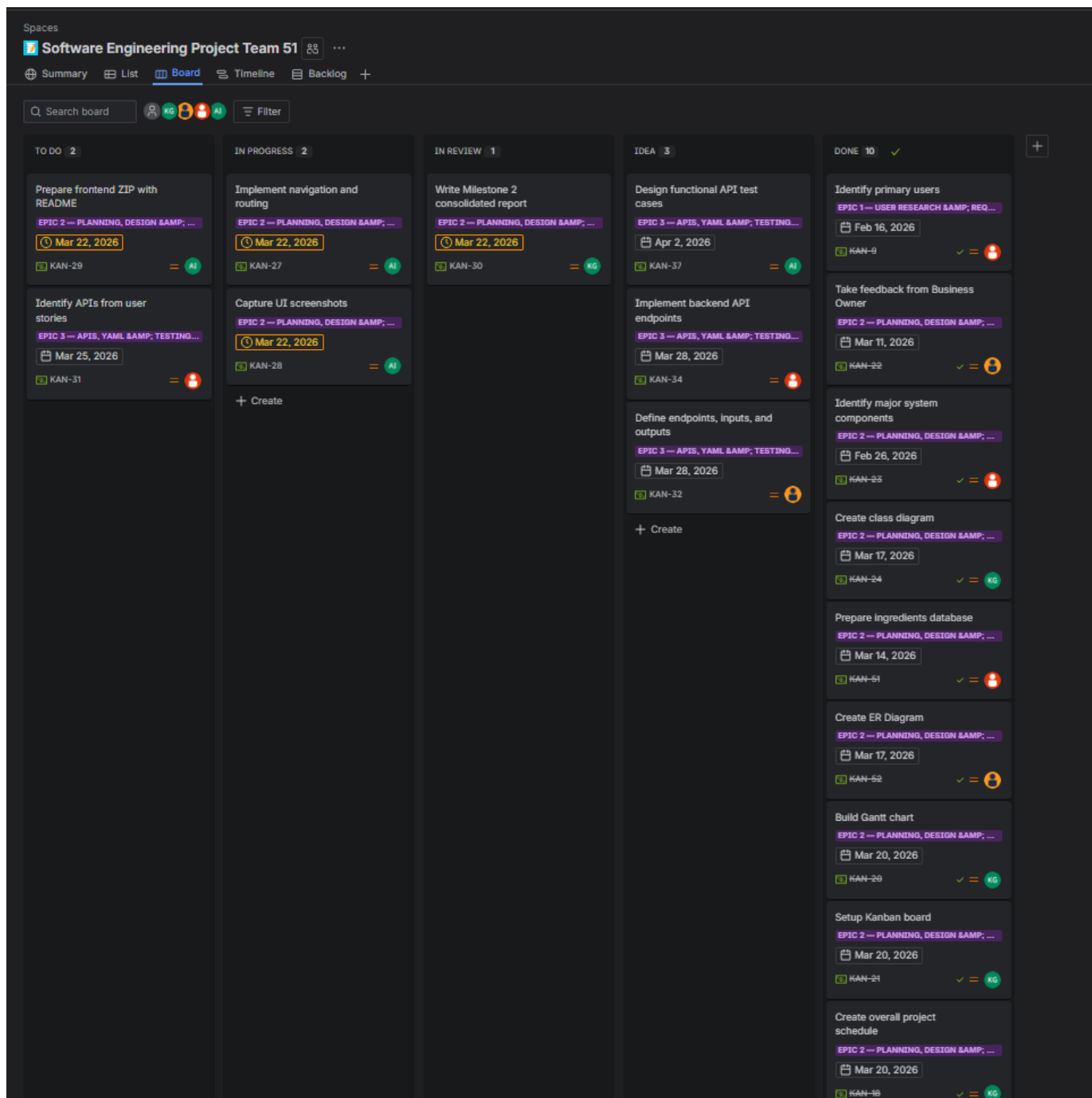


Image 5. Kanban Board

Feedback From the Business owner

During the review of our system design, the business owner provided feedback for different sections of the application to make it more practical and aligned with real-world usage.

Customer Page Feedback:

- The business owner suggested modifying the placement of the “Create Your Own Dish” AI feature. It should be positioned at the top of the menu page so that customers can easily access it. This feature allows customers to describe a dish they want and receive AI-generated suggestions based on available ingredients.
- Another important suggestion was to modify the ordering flow. Instead of directly sending orders to the kitchen, a manager approval step was introduced. Now, when a customer places an order, it first goes to the manager. The manager can approve or reject the order, which helps prevent fake or unnecessary orders (for example, when someone scans a QR code without actually being present). This ensures better control and avoids wastage of ingredients.

Chef Page Feedback:

- For the chef/kitchen section, the business owner suggested improving the design to match the overall system theme with a minimal and clean interface.
- Orders should be displayed in a proper row-based table format for better readability. Additionally, orders should be arranged in a priority queue, where different categories are highlighted using different border or background colors.
- Another key suggestion was to add a “Recipe” button only for customized dish orders. This allows the chef to view the ingredients and preparation steps specifically for AI-generated or custom dishes.

Waiter Page Feedback:

- The business owner suggested simplifying the waiter workflow. When the waiter clicks on a table number, all the corresponding orders should be displayed on a

separate page in a bill-like format. This page should also include a “Pay” button to proceed with payment easily.

Manager Page Feedback:

Several improvements were suggested for the manager section:

- In the recent activity section, low stock notifications should be removed. Instead, low stock items should be highlighted separately in a dedicated category using alert colors or background.
- In the manager settings, three new options should be added to generate QR codes for chef, waiter, and tables.
- For table QR generation, the system should take input for the number of tables and then generate QR codes accordingly.
- In the order management section, the current total cost button should represent the manufacturing cost. The revenue option should be removed, and profit should remain. A new total cost (final cost) option should also be included.
- In menu management, unnecessary fields like “popular type” and “supplier” should be removed. While adding a dish, an image upload option should replace the emoji selection.
- In the pricing section, dishes should be displayed in a row-wise format with a search option. Each dish should have a dropdown to view its ingredients.

Screen Shots of Frontend Pages

Manager Frontend Page Screenshots

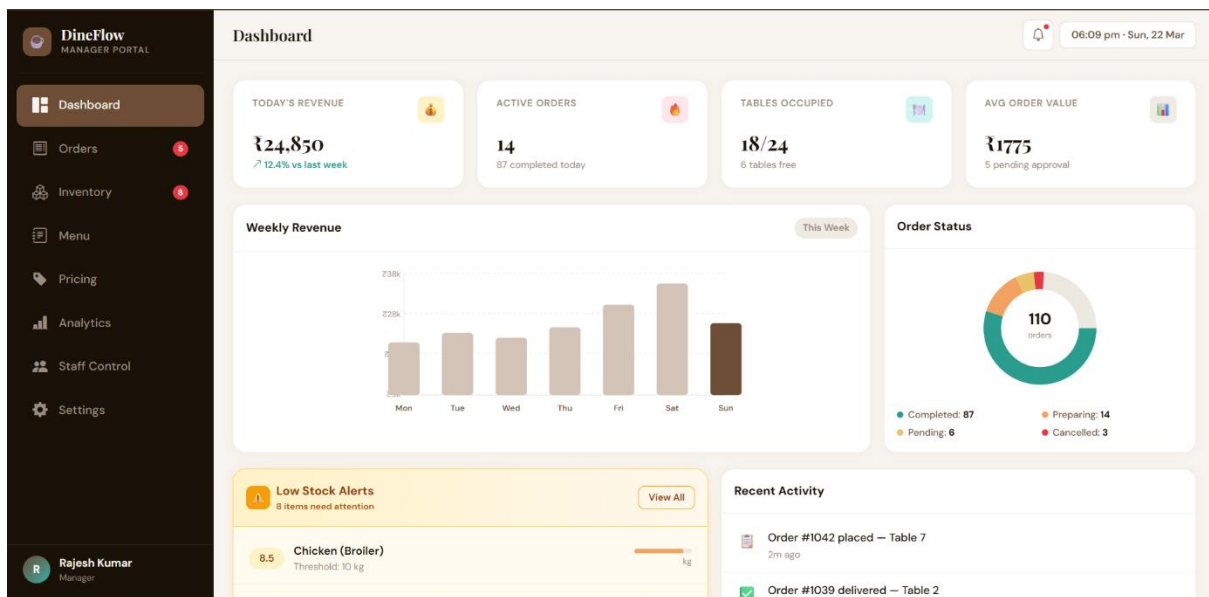


Image 6. Manager Dashboard

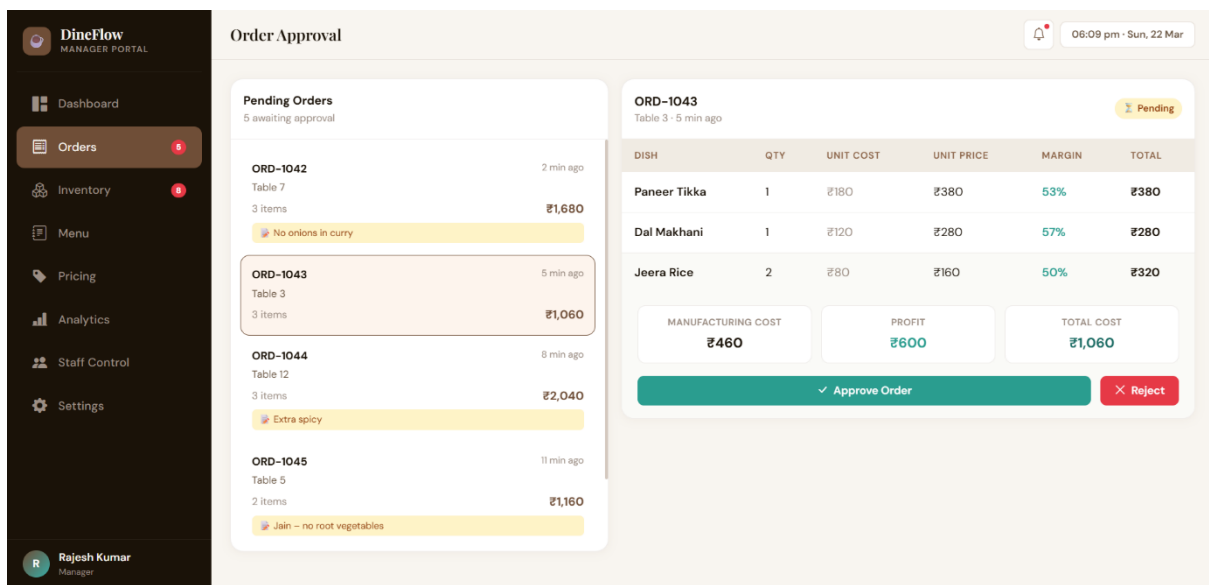


Image 6. Orders Page

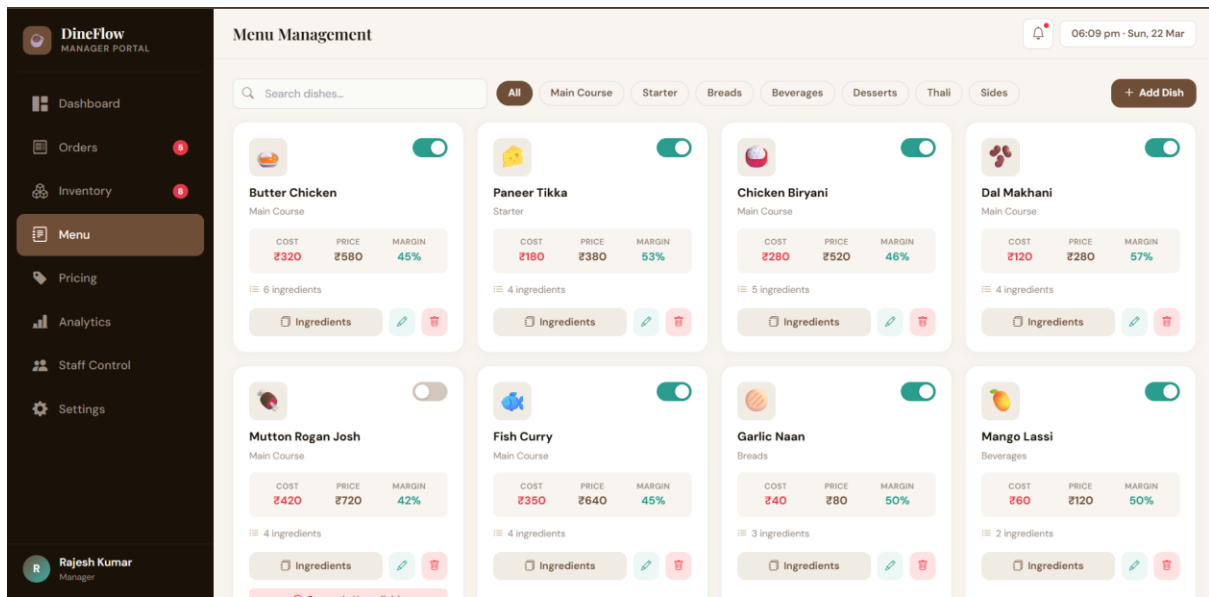


Image7. Menu Page

More Screen Shots of Manager Frontend page can be found here – [Link](#)

Chef Frontend Page screenshots

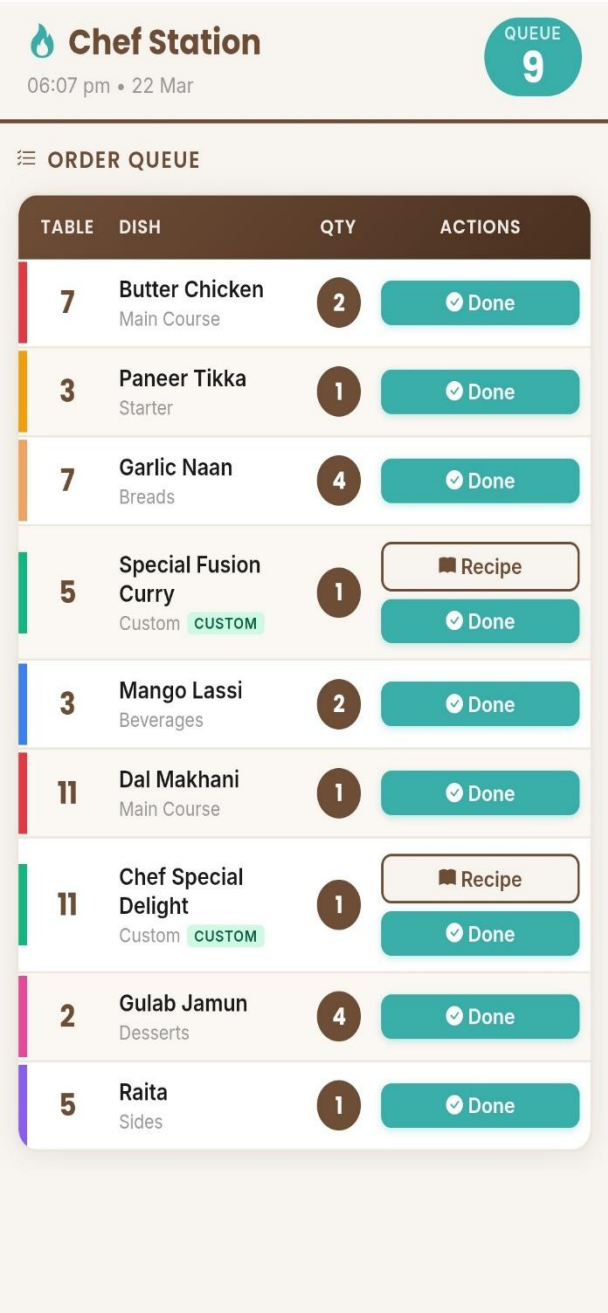


Image 8. Chef Dashboard Page



Image9. Recipe page

Waiter Frontend Page Screen Shots



Image 10. Waiter Dashboard Page

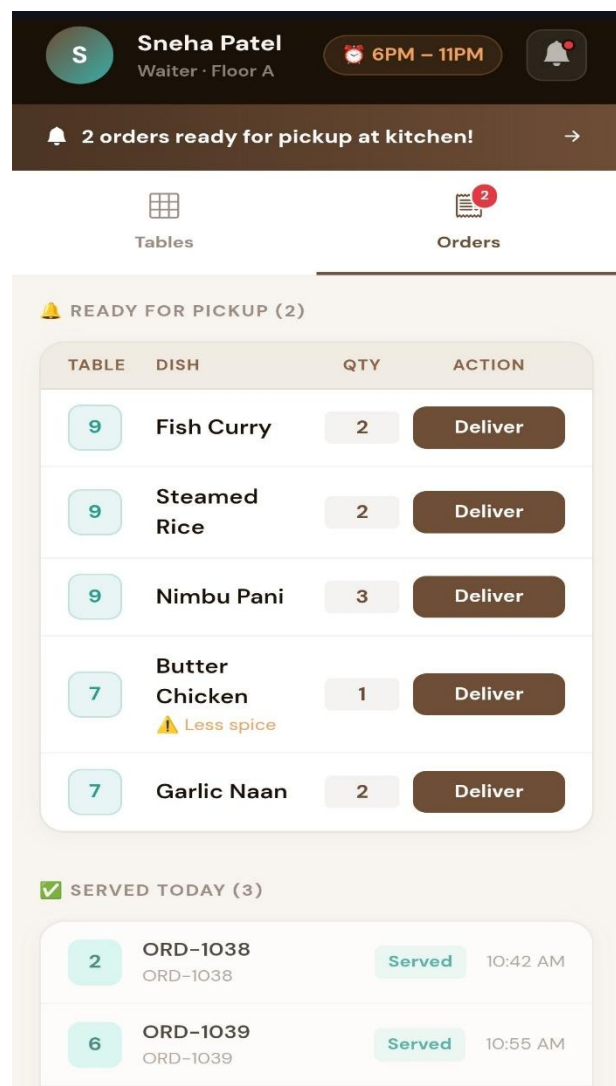


Image 11. Orders Page

More Screen Shots of Waiter Frontend page can be found here – [Link](#)

Customer Frontend Page Screen Shots

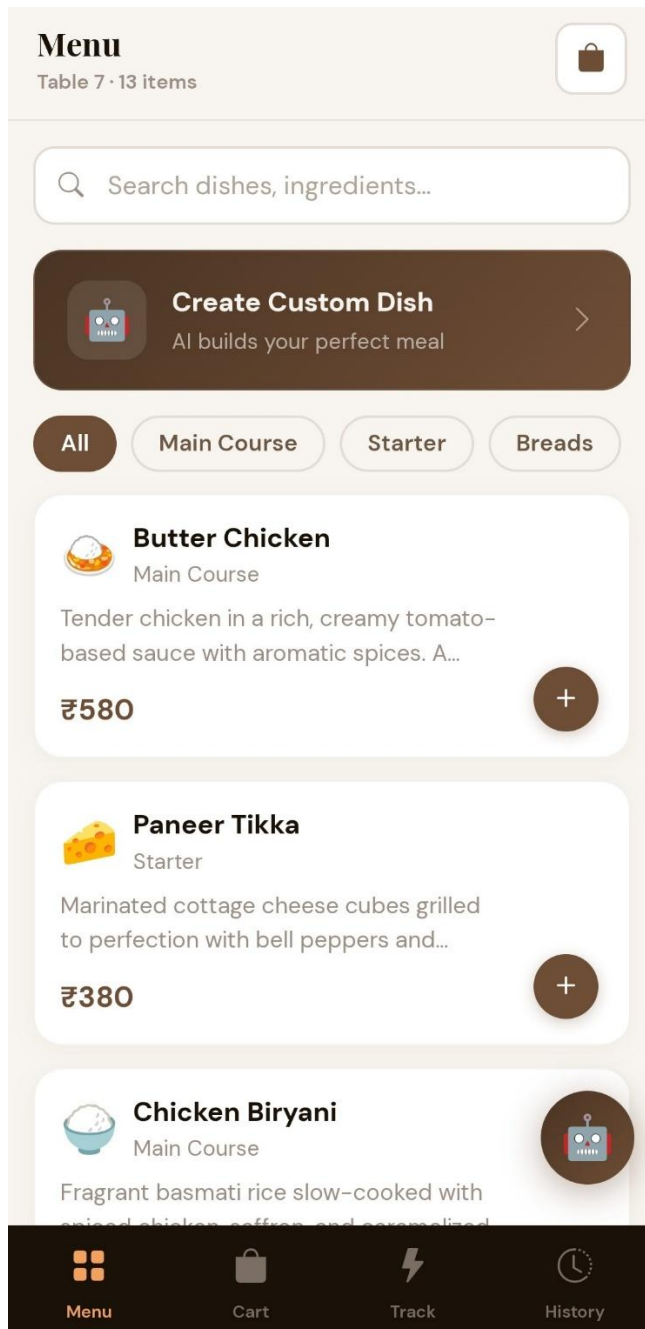


Image 12. Customer Dashboard Page

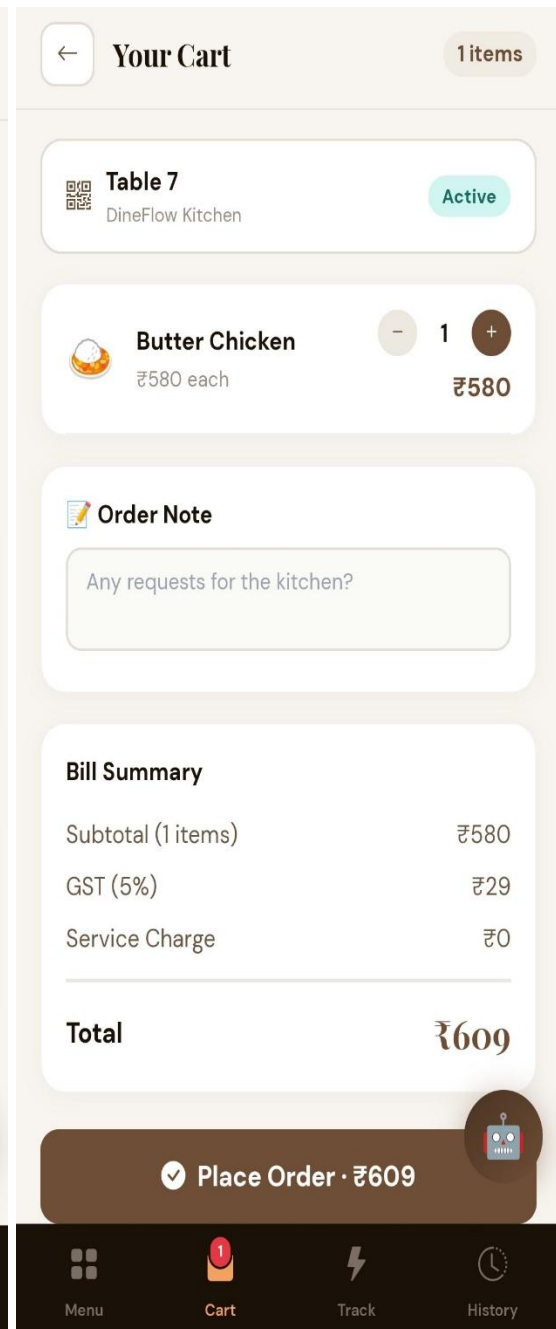


Image 13. Cart Page

More Screen Shots of Customer Frontend page can be found here – [Link](#)

Zip of Frontend code can be found here - [Link](#)